

CSC311 Introduction to Machine Learning Week 8 Exercises

When submitting problems, please **clearly indicate which question you are submitting**. Use the full question number shown below.

Training

W8-T1. **Learning** We wish to build a Naive Bayes classifier to determine whether an email is “urgent”, “normal”, or “spam”. The labelled training set is as follows:

Email	Label
Review document by today	Urgent
Review document	Normal
Account review	Normal
Reset account today	Spam
Account reset review	Spam

- (a) We will use the bag-of-words features as our input features for this model, with the word features ordered alphabetically. Write down the data matrix $\mathbf{X}$.
- (b) We will use a one-hot representation for our targets. Write down the 5×3 matrix $\mathbf{T}$ consisting of our targets. We will use $\mathbf{t} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ or $c = 1$ to represent an “urgent” email, $\mathbf{t} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ or $c = 2$ to represent a “normal” email, and $\mathbf{t} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ or $c = 3$ to represent a “spam” email.
- (c) Explain why we need two parameters to represent $p(c)$. We will call these parameters π_1 and π_2 .
- (d) What does the notation $p(\mathbf{x}|c)$ represent? What about $p(\mathbf{x}, c)$? What about $p(c|\mathbf{x})$? And $p(\mathbf{x})$? What about $p(c)$? What about $p(x_1)$?
- (e) What assumptions (if any) are we making when we write

$$p(\mathbf{x}, c) = p(c)p(x_1|c)p(x_2|c, x_1)p(x_3|c, x_1, x_2) \dots p(x_D|c, x_1, \dots, x_{D-1})?$$

- (f) What assumptions (if any) are we making when we write

$$p(\mathbf{x}, c) = p(c)p(x_1|c)p(x_2|c)p(x_3|c) \dots p(x_D|c)?$$

- (g) Explain why it makes sense, in this problem, to model the conditional distribution $p(x_j|c = 1)$ using a Bernoulli distribution. We will use θ_{j1} to represent the parameter of this distribution, so that in our model, $P(x_j = 1|c = 1) = \theta_{j1}$.
- (h) Similar to above, explain why it makes sense, in this problem, to model the conditional distribution $p(x_j|c = k)$ for other values of k using a Bernoulli distribution. We will use θ_{jk} to represent the parameter of this distribution, so that in our model, $P(x_j = 1|c = k) = \theta_{jk}$.
- (i) Is it true that $\theta_{11} + \theta_{12} + \theta_{13} = 1$? Explain why or why not.
- (j) Use the above data to fit the parameters π_1 and π_2 using maximum likelihood.
- (k) Use the above data to fit the parameters θ_{12} using maximum likelihood, where word $i = 1$ is the word “account”.

W8-T2. **MAP.** From the previous question, how would your estimate for θ_{12} change if we were to use MAP estimation instead of maximum likelihood, with Beta priors and $a = b = 2$?

W8-T3. **Inference** Suppose that we have the following MLE estimates of a Naive Bayes model that classifies whether an email is spam ($c = 1$) or not-spam ($c = 0$). Unlike the problem above, this problem is a binary classification problem.

$$\begin{aligned}
 P(c = 1) &= \pi = 0.15 \\
 P(x_1 = 1|c = 1) &= \theta_{11} = 0.53 \\
 P(x_2 = 1|c = 1) &= \theta_{21} = 0.63 \\
 P(x_3 = 1|c = 1) &= \theta_{31} = 0.23 \\
 P(x_4 = 1|c = 1) &= \theta_{41} = 0.27 \\
 P(x_1 = 1|c = 0) &= \theta_{10} = 0.13 \\
 P(x_2 = 1|c = 0) &= \theta_{20} = 0.33 \\
 P(x_3 = 1|c = 0) &= \theta_{30} = 0.84 \\
 P(x_4 = 1|c = 0) &= \theta_{40} = 0.11
 \end{aligned}$$

Where x_1, x_2, x_3, x_4 are indicators for whether the words “it”, “now”, “buy”, “spam” are in the email. Thus, $P(x_3 = 1)$ is the probability that the word “buy” is in the email.

- Produce a probability estimate for whether a new email consisting of the words “buy it” is spam.
- Produce a probability estimate for whether an email is spam if you knew *nothing* about the content of the email.
- Produce a probability estimate for whether an email is spam if you knew only that the word “spam” is *not* in the email (and nothing about other words).
- Produce a probability estimate for whether the word “buy” is in any email.

W8-T4. **Derivation** The following shows that the log-likelihood of a Naive Bayes classifier decomposes. Annotate what property or assumption is used in each line of the derivation.

$$\ell(\theta) = \sum_{i=1}^N \log p(c^{(i)}, \mathbf{x}^{(i)}) \quad (1)$$

$$= \sum_{i=1}^N \log \left\{ p(\mathbf{x}^{(i)}|c^{(i)})p(c^{(i)}) \right\} \quad (2)$$

$$= \sum_{i=1}^N \log \left\{ p(c^{(i)}) \prod_{j=1}^D p(x_j^{(i)}|c^{(i)}) \right\} \quad (3)$$

$$= \sum_{i=1}^N \left[\log p(c^{(i)}) + \sum_{j=1}^D \log p(x_j^{(i)}|c^{(i)}) \right] \quad (4)$$

$$= \sum_{i=1}^N \log p(c^{(i)}) + \sum_{j=1}^D \sum_{i=1}^N \log p(x_j^{(i)}|c^{(i)}) \quad (5)$$

W8-T5. What is the difference between Maximum Likelihood Estimation and Maximum A-Posterior (MAP) Estimation? What is the main advantage of MAP estimation compared to maximum likelihood estimation?

Validation

W8-V1. During the lecture and the Naive Bayes lab, we classify whether movie reviews were positive or negative using the Naive Bayes model:

$$p(c, x_1, \dots, x_D) = p(c)p(x_1|c) \cdots p(x_D|c)$$

Where c is the label class (positive or negative), and each $x_j \in \{0, 1\}$ represents whether word j appears in the review. We will fit parameters π and θ_{jc} , with $p(c = 1) = \pi$, representing the probability of positive class and $p(x_j = 1|c) = \theta_{jc}$, representing the probability of word j appearing in a review with class c . Suppose that we would like to find the word whose **presence** most strongly indicate that the review is positive. In other words, we wish to find the word j with the largest value of $p(c = 1|x_j = 1)$. Express the quantity $p(c = 1|x_j = 1)$ in terms of the fit parameters θ_{jc} and π .

W8-V2. Using the model in training question 3, produce a probability estimate for whether an email contains the word “buy” if we know that the email contains the word “spam”.

W8-V3. Consider a Naïve Bayes classifier for modelling heart attack y in a patient, based on the presence or absence of chest pain x . $y \in \{0, 1\}$ where $y = 1$ indicates that the patient has suffered a heart attack. $y \sim \text{Bernoulli}(\pi)$. $x \in \{0, 1\}$ where $x = 1$ indicates the presence of chest pain. $x|y = 1 \sim \text{Bernoulli}(\theta_1)$, and $x|y = 0 \sim \text{Bernoulli}(\theta_0)$.

Show that the decision boundary can be expressed as $wx + b$.

Clearly indicate the values for w and b .

Hint Recall that the decision boundary is the points for which the output layer of the classifier is ambiguous i.e., $P(y = 0|x) = P(y = 1|x)$.

W8-V4. Consider the problem of classifying positive and negative reviews discussed in lecture: we are given a dataset $\mathcal{D} = \langle \mathbf{x}^{(1)}, y^{(1)} \rangle, \langle \mathbf{x}^{(2)}, y^{(2)} \rangle, \dots, \langle \mathbf{x}^{(N)}, y^{(N)} \rangle$. Same as we did in lecture, y is a binary variable, that indicates whether the review is negative ($Y = 0$) or positive ($Y = 1$), such that $Y \sim \text{Bernoulli}(\pi)$. We will also assume that $\forall i; 1 \leq i \leq N, \mathbf{x}^{(i)}$ is a vector of D elements, each corresponding to a word in our vocabulary. However, unlike our example in lecture, we will use X_j to indicate the number of times a word in our vocabulary appears in a review. For example, if the second word ($j = 2$) of our vocabulary is “movie”, then $\mathbf{x}^{(4)} = 6$ indicates that the word “movie” appeared 6 times in the fourth review of our dataset. To model the word frequency in the positive and negative reviews, we will use a Poisson distribution:

- $X_j|Y = 0 \sim \text{Poisson}(\lambda_{j,0})$
- $X_j|Y = 1 \sim \text{Poisson}(\lambda_{j,1})$

Find the maximum likelihood estimates for π , $\lambda_{j,0}$, and $\lambda_{j,1}$.

Hint $P(X_j = \mathbf{x}_j^{(i)}|Y = 0) = \frac{\lambda_{j,0}^{\mathbf{x}_j^{(i)}} e^{-\lambda_{j,0}}}{\mathbf{x}_j^{(i)}!}$, where e is Euler’s number.

Generalization

W8-G1. Calculate the decision boundary for the review classification example discussed in lecture. Recall that y is a binary variable, modelled by $\text{Bernoulli}(\pi)$, that indicates whether the review is positive ($y = 1$), or negative ($y = 0$). A review $\mathbf{x} \in \mathbf{R}^D$ is represented using a bag-of words approach, where the j^{th} element in $\mathbf{x}$ indicates the presence ($x_j = 1$) or absence ($x_j = 0$) of the j^{th} word in the vocabulary, such that $x_j|y = 1 \sim \text{Bernoulli}(\theta_{j,1})$, and $x_j|y = 0 \sim \text{Bernoulli}(\theta_{j,0})$.

Show that the decision boundary can be expressed as $\sum_{j=1}^D w_j x_j + b_j$.

Clearly indicate the values for w_j and b_j .

W8-G2. Remember the Kullback-Leibler (KL) divergence from W2-G3, that measures the dissimilarity between distributions. Let $P_{\text{data}}(\mathbf{x})$ be the true data distribution. Learning the data distribution is often intractable. We therefore often work with a simpler distribution $P_{\text{model}}(\mathbf{x}; \theta)$, parametrized by θ . In lecture, we discussed the **Maximum Likelihood Estimate** (MLE) that finds the parameters that maximizes the likelihood $P(\mathcal{D}|\theta)$ i.e., $\text{argmax}_{\theta} P_{\text{model}}(\mathcal{D}|\theta)$. Prove that finding the θ that minimizes the KL-divergence $KL(P_{\text{data}}(\mathbf{x})||P_{\text{model}}(\mathbf{x}; \theta)) = \mathbb{E}_{\mathbf{x} \sim P_{\text{data}}} \left[\log \frac{P_{\text{data}}(\mathbf{x})}{P_{\text{model}}(\mathbf{x}; \theta)} \right]$ corresponds to finding the maximum likelihood estimate for $P_{\text{model}}(\mathbf{x}; \theta)$.

Hint: You might find the law of large numbers helpful for solving this problem.